Alexandria University
Faculty of Veterinary Medicine
Anatomy & Embryology Department

Avian Anatomy

(External features, skeleton)

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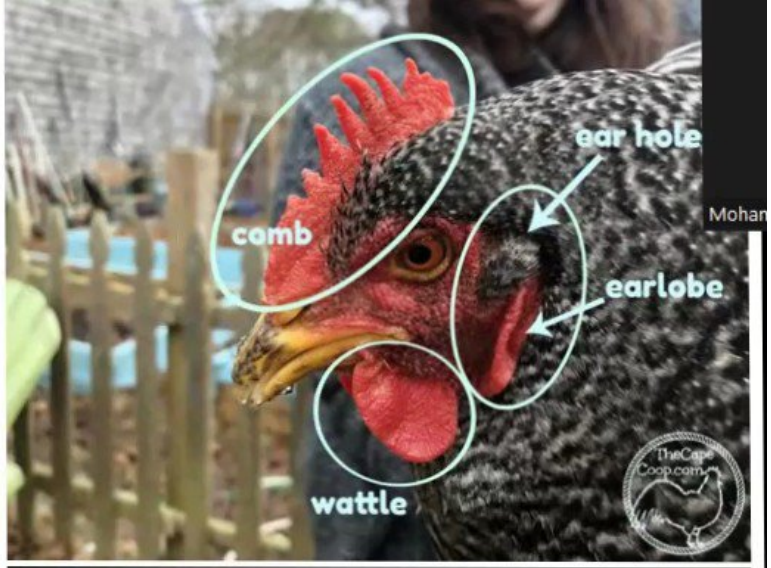
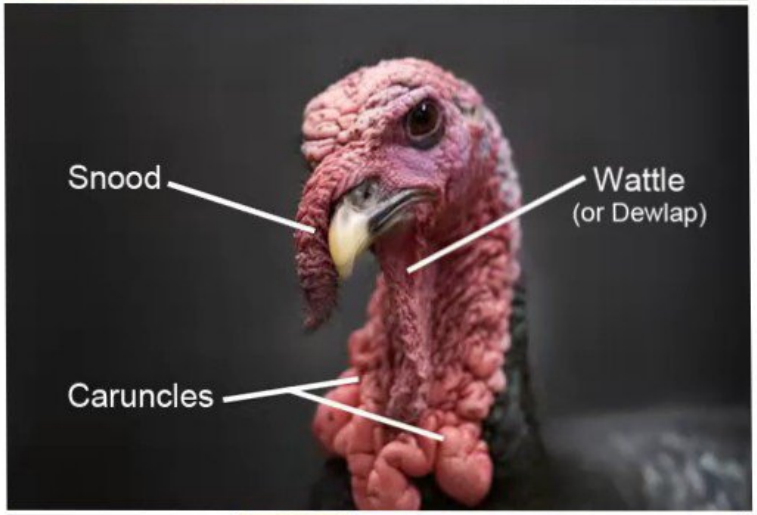
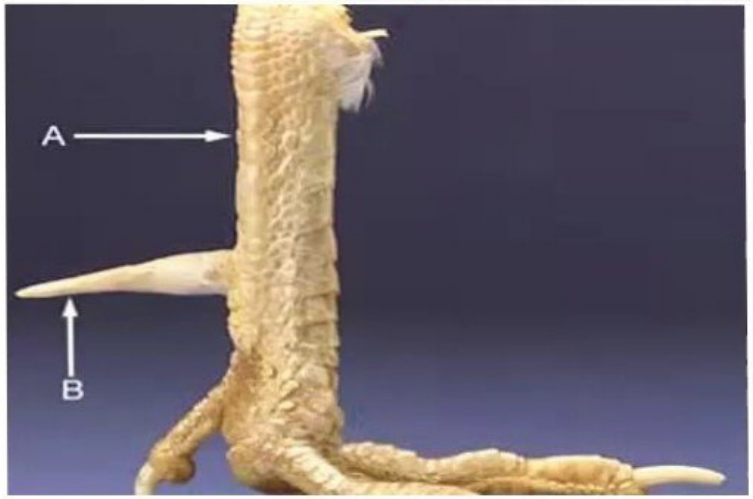
External features



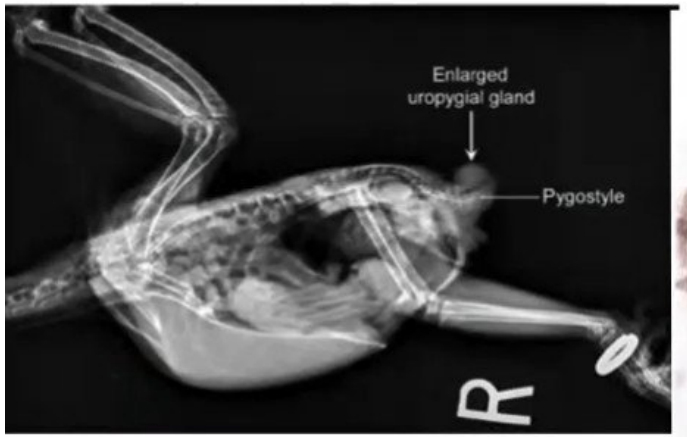


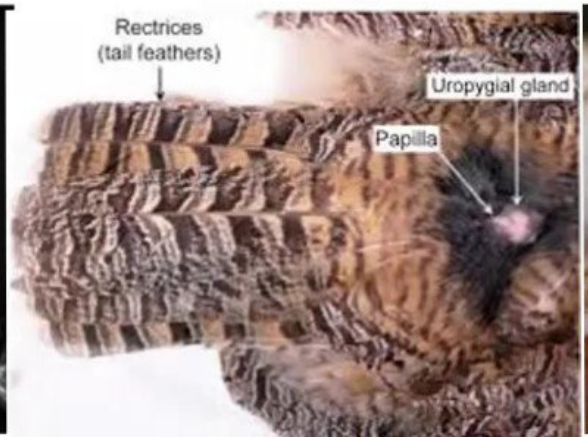


- ***comb, wattles,*** and **ear lobes (snood of turkeys)** are soft ornamental outgrowths of the skin of head.
- **scales** on the shanks and feet are cornified epidermal patches similar to those of reptiles.
- ***spur*** developed on the caudomedial surface of the rooster's shank is used as a weapon.
- three discrete skin glands: **sebaceous *uropygial,* aural, and vent glands.**
- **absence of sweat glands** means that the birds must lose heat through their skin and evaporation from the respiratory system.

Uropygial Gland (Oil Gland)

- The uropygial gland is an epidermal gland in birds, located on the dorsal base of the tail , that produces a waxy oily secretion plays an important role in helping the hen to keep her feathers waterproof and in good condition.
- secretions of the gland empty onto the skin surface through one or more nipple-like pores. Most birds rubbing their head and beak over the gland pore and then distribute the accumulated oil over the feathers of the body and wings, skin and legs.
- The gland is found in most birds and is best developed in water birds, prominent in budgerigars and African greys but absent from many other parrots (Amazon parrots), ostriches, and many pigeon.





Skeleton of the chicken

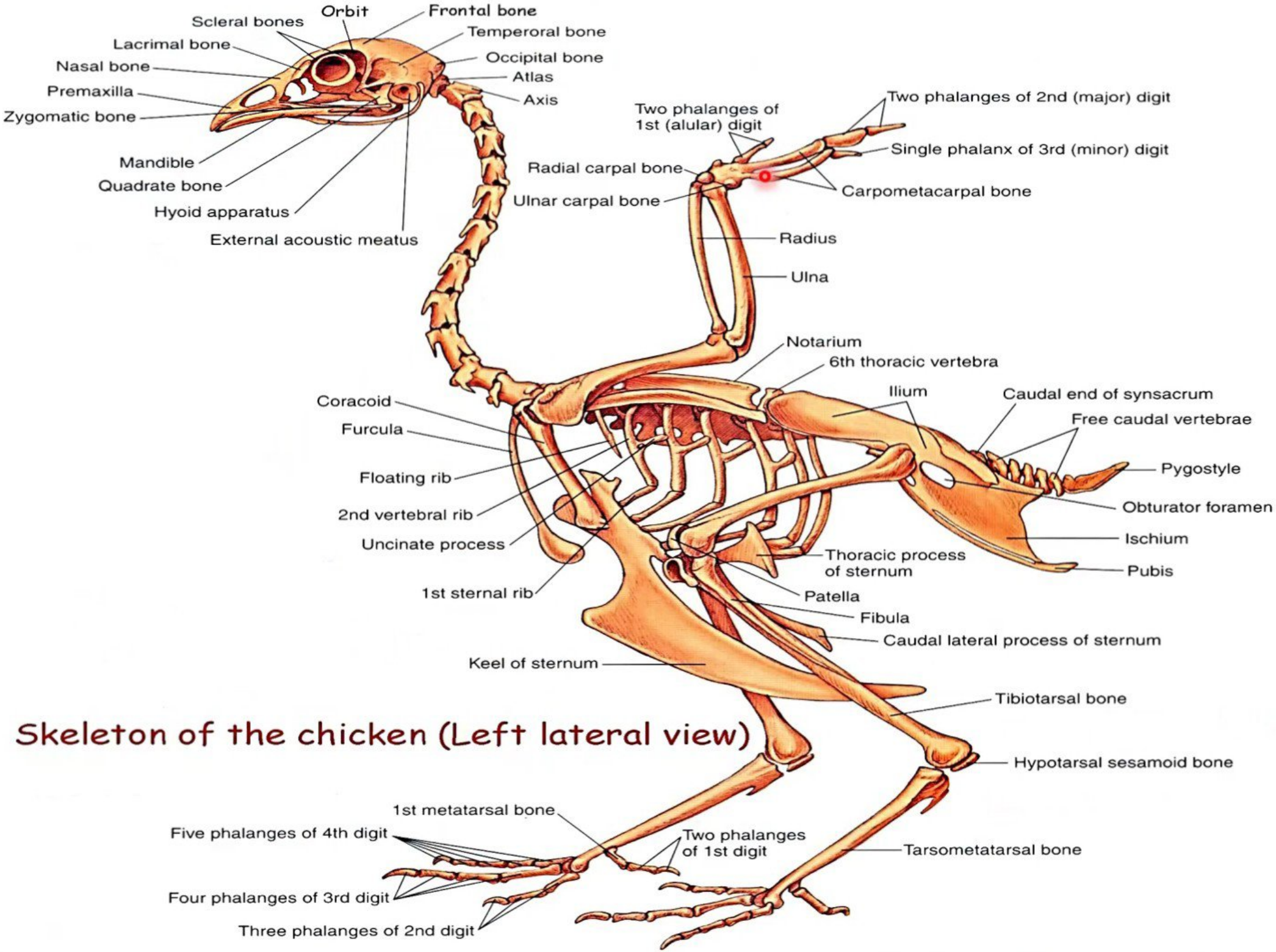
Divided into two parts:

A- Axial skeleton:

1. Skull
2. vertebreal column
3. ribs and sternum

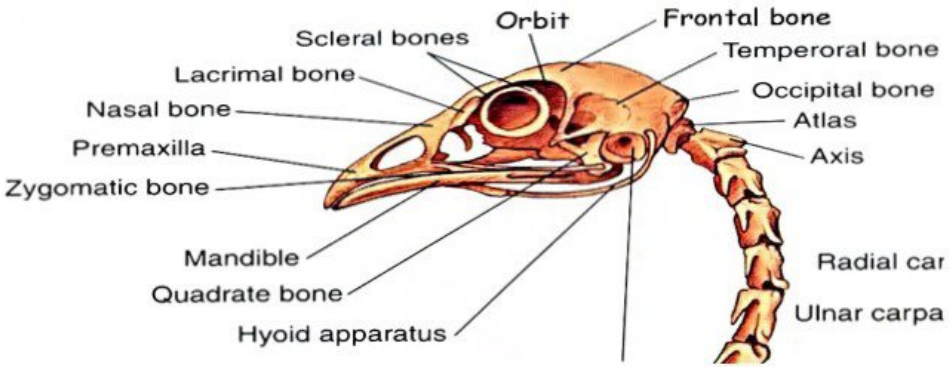
B- Appendicular skeleton:

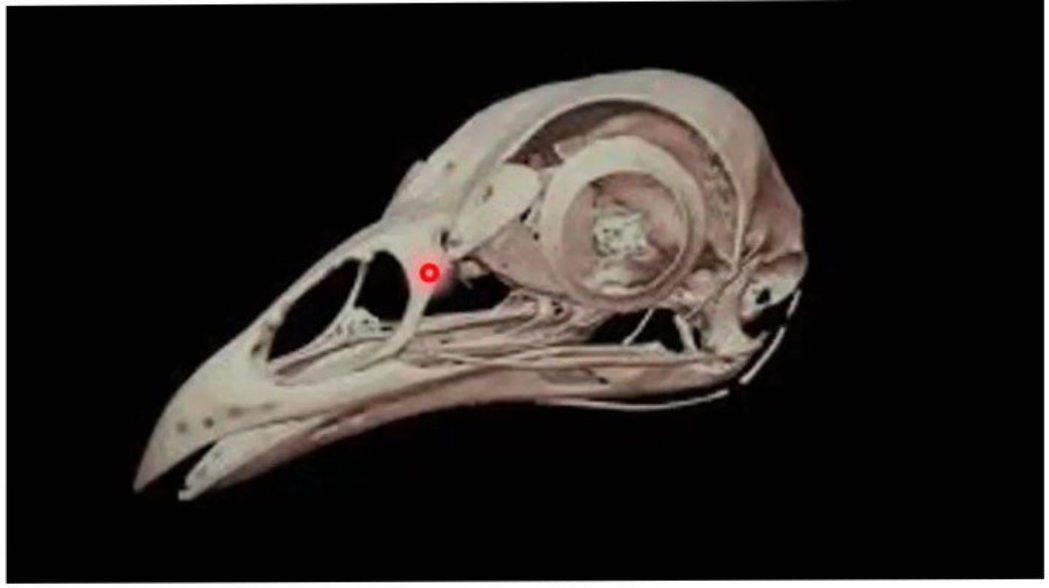
bones of limb (some modifications according to requirement of locomotion).



Skeleton of the chicken (Left lateral view)

Skull





The skull of the chicken shows **two large orbit**, separated by a **thin septum formed sphenoid and ethmoid bones**. Behind the orbit is the **rounded cranium** whose cavity enclosed the brain and is covering.

Cranial group: as mammals in name and number, except interparietal bone is absent

Facial group: as mammals, except there is turbinate bone + quadrate bone which articulate with mandible.

2- Vertebral column

The vertebral formula for chicken (C14 +T7 + L-S14 + Cd6)

A- Cervical vertebra:(14 in fowl and duck and 17 in goose)

Atlas (1st), axis (2nd)

Remaining c.v. has (body+ arch+ processes)

B- Thoracic vertebra (7): (7 in fowl and 9 in duck and goose)

- Well-developed dorsal spinous processes except last have a very clear ventral spinous process
  - 2- 5 th.v.: fused notarium
  - 1 + 6 th.v.: free

C- Lumbo-sacral region (14)

- 7 th.v. fused with lumbosacral mass+ 1st coccygeal v. form: synsacrum

D- Caudal vertebra

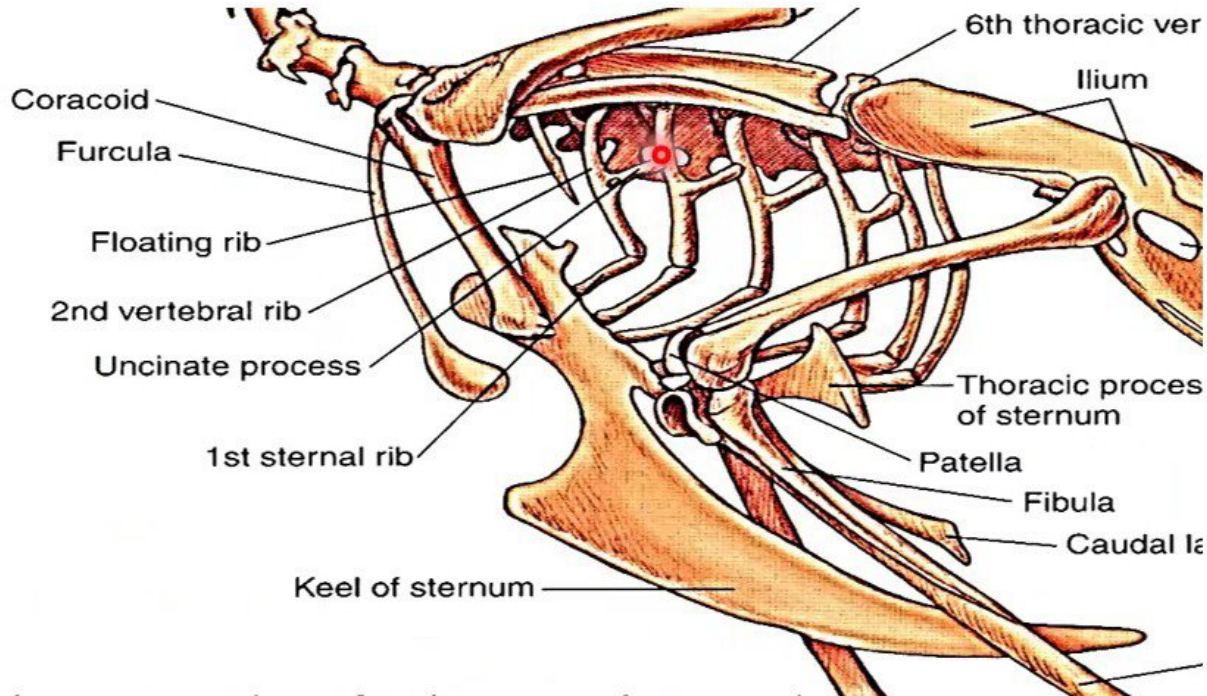
- 4-9 or 10 caudal vertebra
- Final part of vertebreal column is formed by fusion of 4- 8 caudal vertebra called pygostyle (plowshare bone) which paly important role in flight. Pigeon: pygostyle: fusion of 7 to 10 cd. Vert.

3- Ribs

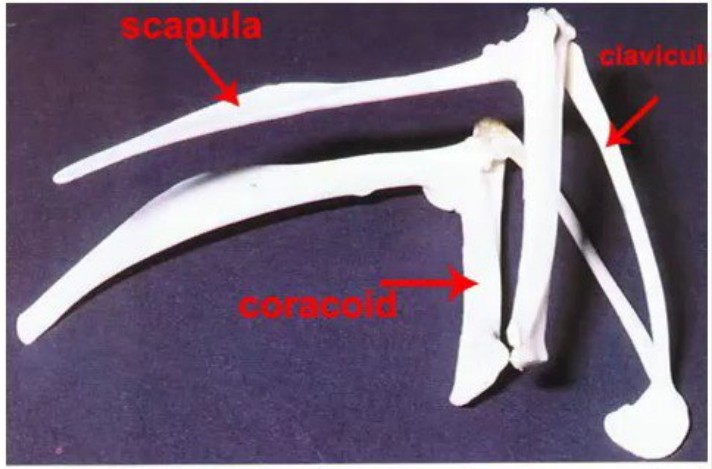
- 7 pair of true ribs
- consists of: (vertebral rib+ sternal ribs
- No costal cartilage
- Vertebral ribs except 1st and last: bear on the caudal border, flattened processes which overlap the next ribs.

4- Sternum (breast bone) (keel bone)

Form the floor of thoracic cavity + great part of abdominal cavity. It is quadrilateral



Appendicular skeleton



Forelimb (wing)

- **Pectoral girdle:** 3 pair of bones

Fused clavicles, Coracoid, and Scapula.

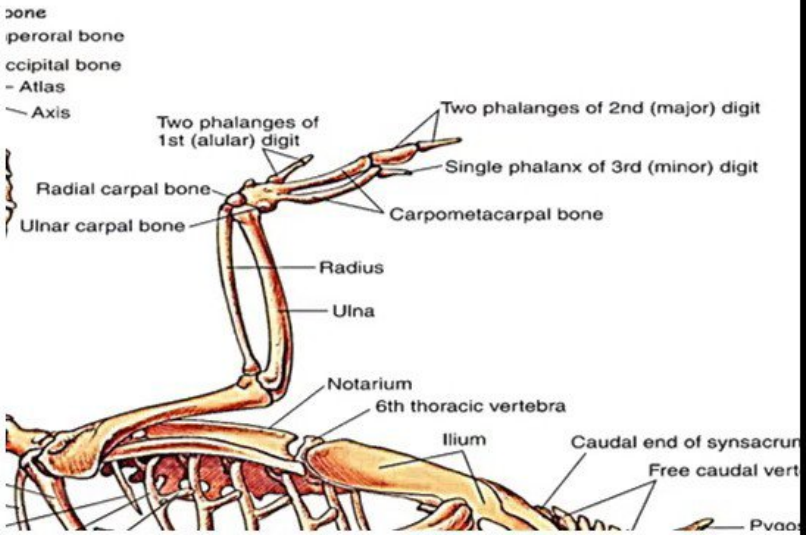
- **Humerus:** articulate proximally with scapula + coracoid
- **Radius and ulna:** (radius smaller than ulna)
- **Carpus:** two row proximal and distal
- **Metacarpus:** consists of 3 bones form single bone

Digit: 3 in number 1st, 2nd and 3rd digit.

The 1st digit is small (2 phalanges).

The 2nd digit is largest (2 phalanges).

The 3rd digit is smallest (1 phalanx).



Pelvic limb

❖**Pelvic girdle:**

✓os coxae (ilium+ischium+pubis) not meet ventrally like mammals.

❖**Femur** It articulate with acetabulum proximally and fibula and patella distally

❖**Tibia:** longer bone of leg

❖**Tarsus:** two rows

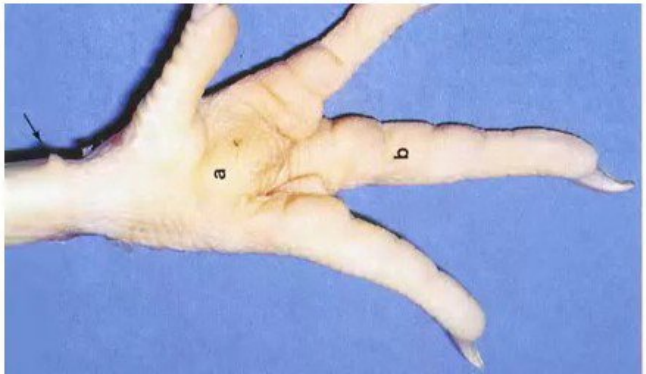
❖**Metatarsus:** consists of 2nd + 3rd + 4th metatarsal

❖**Digit:** 4 in number

1st, 2nd 3rd and 4th digit

1st formed of 2 phalanges.

2nd formed of 3 phalanges.

3rd formed of 4 phalanges.

4th formed of 5 phalanges.